A0 INVESTIGATION LAYER (PLANNED)

Design rules

- **Cite or drop:** every statement points to a frame, log line or record, or it is not reported
- **Corroborate:** a finding is confirmed only when two independent sources agree
- **One clock:** all sources in UTC; each device's clock offset is estimated and shown
- **Abstain:** UNKNOWN stays UNKNOWN; the report never guesses
- **Reproduce:** same evidence and config hash give the same report

The detection engine underneath is built and measured (slides 5–7); this layer is planned work.

PLAN - from a network alert to a reconstructed, court-ready case (20 Nov PoC)

Each row is one evaluation criterion. Left: what is built and measured today. Right: what the 20 Nov PoC adds (planned, not yet measured).

Criterion · built today	Added by the 20 Nov PoC (planned)
Evidence processing & extraction Today: PCAP to directional flows, SHA-256 on intake	Add Zeek/Suricata, Sysmon, firewall, proxy and DHCP logs plus offline threat-intel lists, each hashed and logged on intake
Cross-source correlation Today: four engines on one capture	Link IP, hostname, user and process across sources; a finding is "confirmed" only when two independent sources agree
Timeline & reconstruction Today: per-host trajectory, Normal to High	One merged UTC case timeline with per-device clock-offset correction; attack stages tagged with MITRE ATT&CK IDs
Traceability & explainability Today: reasons per incident, UNKNOWN, hash ledger	Every finding cites frame numbers, flow IDs, a Wireshark filter and a sealed excerpt PCAP; BSA Sec. 63 certificate draft
Multimedia / fraud analysis Today: not covered	Files recovered from cleartext traffic (hash, true type, entropy), image-tamper checks, look-alike and generated-domain scoring
Performance & usability Today: 22,011 packets/s; 5 min of capture in 1.85 s	Case dashboard (timeline, graph, findings with "why"), one-click PDF report

A1 RUBRIC COVERAGE

APPENDIX - each criterion mapped to slides, status and evidence

Criterion		Slides	Status in this prototype	Evidence
Evidence processing & extraction		5, A2	**Implemented:** PCAP input only; directional 5-tuple flows; 5 s windows, 1 s step, per host; SHA-256 on intake	demo1: 27,149 packets → 3,527 flows → 6,600 host-windows in 1.77 s
Cross-source evidence correlation		3–5, A0, A2	**Partial:** four engines must agree on one capture; these are four analyses of the same data, not independent sources. Log sources planned (A0)	Ablation: without the graph engine 16/16 → 4/16
Crime timeline & reconstruction		4–5, A3	**Partial:** per-host cyber-incident (threat-trajectory) reconstruction; no multi-host case timeline yet	demo1: first engine 135 s, 2-of-4 at 144 s, incident at 146 s
Accuracy, traceability & explainability		6–7, A2, A4	**Implemented and measured (synthetic):** written reasons, frozen thresholds, UNKNOWN rejection, hash-linked ledger; frame-level citation planned	16/16 detected; 8/14 named, 6 UNKNOWN; 5/5 never-seen UNKNOWN
Multimedia / fraud analysis		8, A0, A5	**Partially supported:** flow metadata only; smallest credible extension proposed as future work (A5)	Meta Data Extraction capability (elaborated on A5)
Performance & usability		6–7, A4	**Measured:** throughput, latency, memory on one CPU core; dashboard exists; no user study yet	22,011 pkt/s, 162 Mbps, 549 MB
Presentation & demonstration		3–9, A0–A5	**Provided:** worked example, reconstruction, limitations, reproducible demo steps, 3-minute demo video	Commands and expected outputs on A5

* *Implemented = built and run on held-out synthetic captures; Partial = narrower than the criterion; Planned/Not supported = not built.*

A2

WORKED EXAMPLE: ONE INCIDENT

APPENDIX - demo1_c2 traced from the capture to the sealed record (all values from results/)

1 Input

- demo1_c2.pcap, 300 s, synthetic held-out capture
- SHA-256 05615006…bd6bc928b
- 27,149 packets, read-only, nothing sent

2 Extraction

- 3,527 directional 5-tuple flows
- 6,600 host-windows (5 s window, 1 s step)
- processed in 1.77 s on one CPU core

3 Four engine scores, host 10.0.0.25

- Mahalanobis 0.20 · Isolation Forest 0.33 (weak alone)
- TCN 1.00 · Graph 0.97 (strong)
- same capture, four different representations

4 Fusion and agreement

- $S = 1 - \Pi(1 - A_i)^{0.25}$ -> peak 0.9998 ≥ τ_sev 0.985
- C = #engines with $A_i \geq 0.99$ -> 2 ≥ n_min 2
- persistence P = 0.99 over k = 6 windows

5 Attack-vector decision

- $S_C2 = (C/4)\cdot P\cdot \Sigma\, w_j\phi_j = (1.97/4)(0.99)(0.943) = 0.463$
- RECON 0.145 · EXFIL 0.072 · DDOS 0.000
- 0.463 ≥ τ 0.415 and margin 0.318 ≥ δ 0.095 -> C2

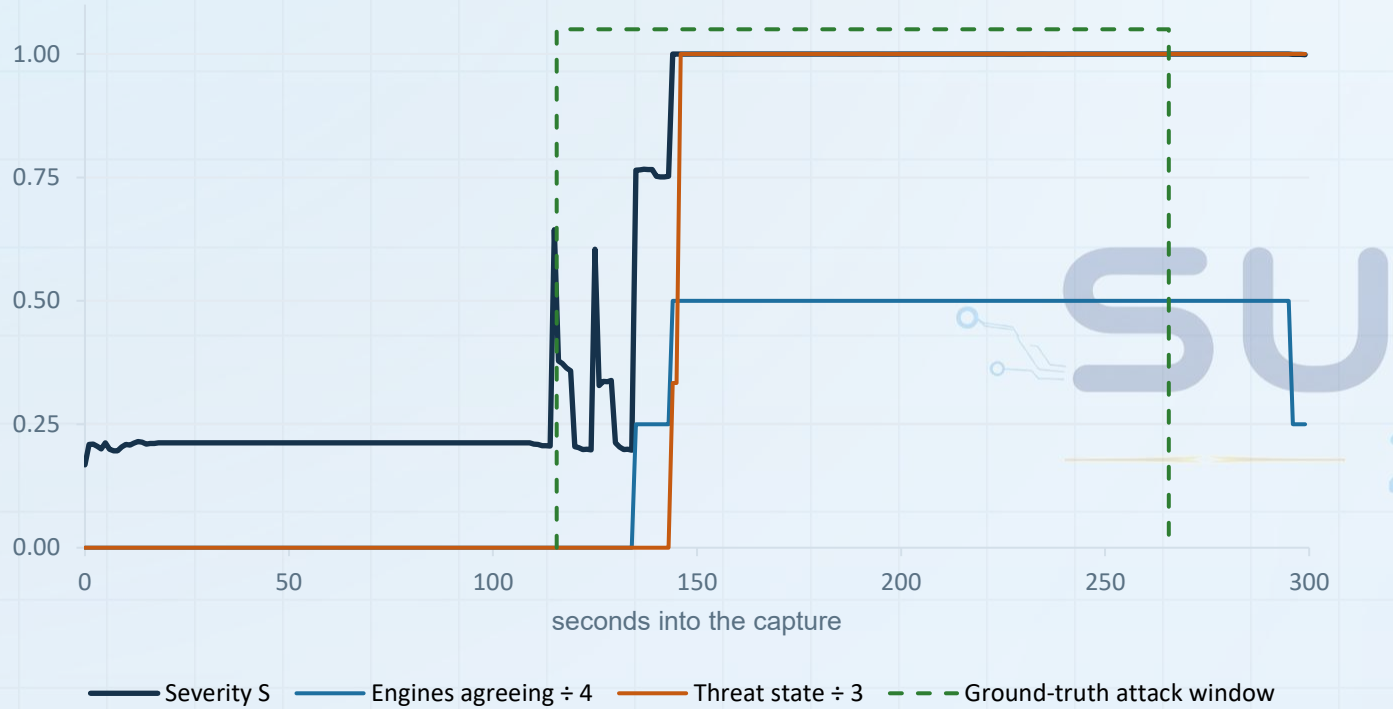
6 Reasons and record

- destination focus 1.00 · periodicity 0.89
- destination persistence 0.87 · jitter regularity 0.88
- ledger #0 prev 0000… hash b62f92db2a3f…

Recomputing S_C2 from the listed φ values and the frozen weights in configs/default.yaml gives 0.4626, the reported score. Ground truth (test only): beaconing 116 - 266 s; incident window 146 - 299 s. Agreement here is between analyses of one capture, not independent evidence sources.

INCIDENT RECONSTRUCTION

APPENDIX - cyber-incident (threat-trajectory) reconstruction for demo1_c2, one host, one capture



Time	Reconstructed event	Evidence
0 - 114 s	Baseline: severity ≤ 0.21, no engine past its threshold	per-host windows
115.6 s	Beaconing to 198.51.100.42 begins (unknown to the system)	ground truth (test only)
135 s	First engine crosses its threshold (C = 1)	engine scores
144 s	Second engine agrees (C = 2): state Emerging	fusion record
146 s	Persistent, then High-confidence: incident opened and sealed	ledger #0
155 s	Peak severity 0.9998	fusion record
265.6 s	Beaconing ends (ground truth)	ground truth (test only)
299 s	Incident still open at end of capture; labelled C2-like beaconing	incident report

Drill-down today: ledger record -> incident window -> host-windows -> flows. Frame-level citations and a merged multi-host, multi-source case timeline are planned (A0). Limitation shown here: the incident stays open after the beaconing stops.

A4 VALIDATION AND LIMITATIONS

APPENDIX - synthetic evaluation: how it was tested, what it shows, what it does not

Test design / metric	Result (synthetic held-out captures)
Training data	Benign traffic only; thresholds and config hashed before any test capture was scored
Test set	28 captures: 1.5 h normal + attack, 0.83 h benign look-alikes
Attack episodes detected	16/16 (C2 4/4, Recon 4/4, DDoS 2/2, Exfil 4/4, never-seen 2/2)
Median time to detect	19 s
Known incidents named	8/14: C2 4/4, DDoS 2/2, Exfil 2/4, Recon 0/4; the other 6 left UNKNOWN
Never-seen incidents	5/5 left UNKNOWN; 0 attack types confused
False alerts	9.3/h on normal traffic; 42.0/h on benign look-alikes
Throughput / memory	22,011 packets/s, 162 Mbps, one core, 549 MB; 5-min capture in 1.85 s (median)

Sources: results/ablation.json, classification.json, robustness.json, benchmark.json, FREEZE.txt in the project repository.

What the hash-linked ledger establishes

- Any edit, deletion or reordering of a sealed incident record is detected (5/5 tamper tests)
- With the head hash kept in the case file, re-sealing a forged chain is detected

What it does not establish

- That the original PCAP is authentic or complete, or that a finding is correct
- No distributed consensus yet: one hash chain; PBFT replication is planned

Known limitations

- Synthetic traffic only; CIC-IDS2017 and CTU-13 runs not yet done
- False alerts above the < 1/h target, highest on periodic benign services
- Reconnaissance detected but never named; slowest exfiltration rate tested missed (0/2)
- One capture, one data source; no image, audio or document analysis

A5 REPRODUCIBLE DEMONSTRATION

APPENDIX - what an evaluator runs, and what they should see

#	Run	Expected output / what to observe
1	python scripts/run_demo.py	Five lines: C2->C2, RECON->UNKNOWN, DDOS->DDOS, EXFIL->EXFIL, UNKNOWN->UNKNOWN, each incident with written reasons
2	python scripts/build_dashboard.py, open results/dashboard.html	Overview; incident 'why' panel; temporal view with the attack window; recon graph (21 peers, 15 new links); ledger INTACT
3	python video/ledger_check.py --tamper-test	6 records INTACT; edits, deletion and swap each break the chain: 5/5 detected
4	python scripts/run_ablation.py	Full system 16/16 attack episodes; without the graph engine 4/16
5	python scripts/benchmark.py	About 22,000 packets/s on one CPU core (results/benchmark.json)
6	python -m pytest tests	38 tests pass

Steps 1–3 are shown in the 3-minute demo video submitted with this deck. Steps 1, 2, 4–6 need the full engine source in the repository.

Multimedia / fraud: honest status

Partially supported today. UniAethel analyses flow metadata; it does not directly process images, audio, video or documents.

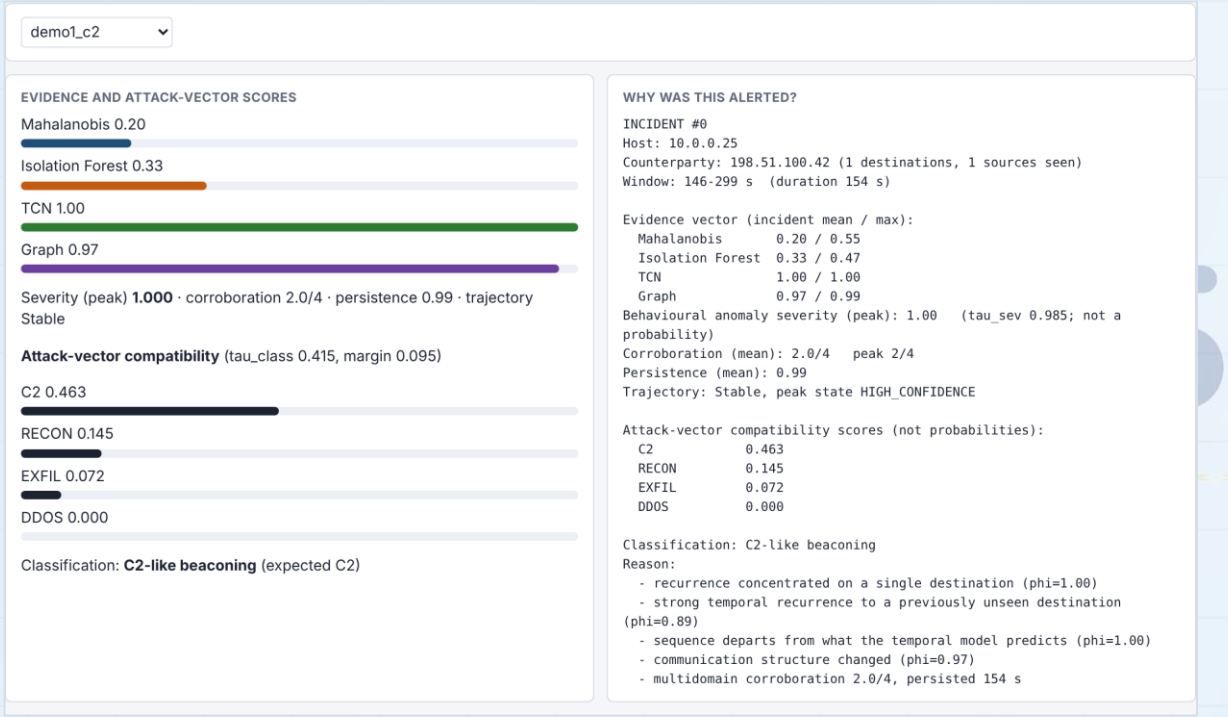
Smallest credible extension (future work)

- Recover files sent over cleartext protocols in the PCAP; record hash, true type vs extension, entropy
- Metadata consistency and error-level checks on recovered images
- Look-alike domain scoring on observed DNS names
- Every result sealed in the same ledger

Needs a labelled set of genuine and altered files to validate.

PROTOTYPE DASHBOARD

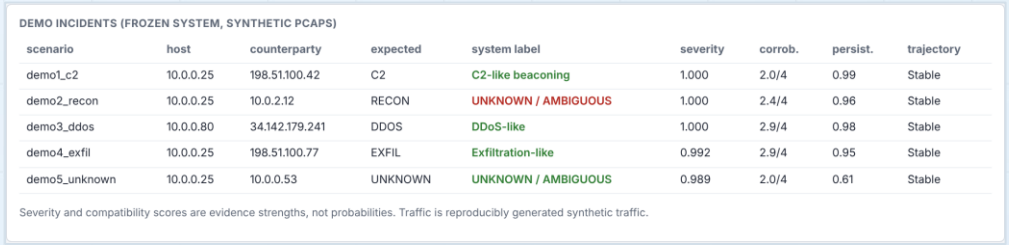
APPENDIX - real screenshots of the working prototype (results/dashboard.html, synthetic demo captures)



Incident detail (demo1_c2): engine and attack-vector scores, and the written reasons for the C2 label



Temporal view: shaded band = real attack window; lower panel = threat state



Overview: five held-out demo captures; recon flagged but left UNKNOWN